Deep Learning

Week 11

Question: 1

While building a part of speech (POS) tagger, you find that the tag output for a word frequently depends on the previous words and their tags.
What aspect of sequence learning does this illustrate?

Options:

- A) Variable length input handling
- B) Fixed size input constraint
- C) Dependence between successive inputs
- D) Independence between inputs

Correct Answer: C

Question: 2

You want to classify a video based on the full sequence of images it contains. The number of frames varies per video.
What challenge does this scenario pose that sequence learning must address?

Options:

- A) Fixed size input requirement
- B) Sequence independence
- C) Processing variable-length sequences with temporal dependencies
- D) Classifying single images independently

Correct Answer: C

Question: 3

Consider the sentence: "The quick brown fox jumps over the lazy dog." You want to predict the part of speech tags for each word. How does the sequence nature affect model design?
Which design principle must be incorporated?

Options:

- A) Treat each word independently
- B) Use a model that captures sequence dependencies among words
- C) Use fixed-length windows ignoring word order
- D) Remove the order and process the sequence as a bag of words

Correct Answer: B

Question: 4

An RNN has hidden state vector s_i at time i. The hidden state at the next time step is computed as $s_i = \sigma(Ux_i + Ws_{i-1} + b)$, where x_i is the input.
What do the matrices U and W represent?

Options:

A) U transforms the current input
B) W transforms the previous hidden state
C) U is the output weight matrix
D) W is the bias term

Correct Answer: A, B

Question: 5

Suppose at time step 3, the RNN hidden state is $s_3 = \sigma(Ux_3 + Ws_2 + b)$. The output at time step 3 is computed as $y_3 = O(V\ S_3 + c)$.
What role does the matrix V play in the RNN?

Options:

A) It transforms input to hidden state
B) It transforms hidden state to output
C) It updates the hidden state recurrently
D) It acts as a bias term

Correct Answer: B

Question: 6

You are building a language model to predict the next word in sentences of varying length.
Why are RNNs preferred over feedforward networks in this scenario?

Options:

A) Because RNNs use fixed-size inputs only
B) Because RNNs can handle sequences of variable lengths due to recurrent connections
C) Because feedforward networks are faster
D) Because RNNs ignore previous outputs

Correct Answer: B

Question: 7

At each time step in a sequence, you want the network to consider both the current input and the information seen in previous time steps.
How does an RNN achieve this?

Options:

- A) By using separate parameters for each time step
- B) By ignoring previous inputs and considering only current input
- C) By using a recurrent connection with a hidden state that carries information forward
- D) By resetting the network state at every time step

Correct Answer: C

Question: 8

You have an RNN unrolled for 4 timesteps processing a sequence. You want to compute the gradients of the loss at the last timestep w.r.t. the recurrent weight matrix W.
How is this gradient computed in BPTT?

Options:

- A) Using only the current timestep's computation
- B) By summing gradients across all timesteps due to the dependency chain
- C) Ignoring previous states and considering only output weights
- D) By computing gradients independently for each timestep without summation

Correct Answer: B

Question: 9

A researcher is using a vanilla RNN to model a sequence of 1000 time steps. She observes that the early inputs have negligible effect on the output at the final time step.
Which of the following is/are the most plausible reason(s) for this behavior?

Options:

A) The RNN suffers from the vanishing gradient problem.
B) The RNN forgets early inputs due to cumulative non-linear transformations.
C) The input sequence is too short for the model to learn dependencies.
D) The RNN lacks a mechanism to selectively retain long-term information.

Correct Answers: A, B, D

Question: 10

You train an RNN using truncated BPTT over sequences of length 100. The truncation window is 20. You notice that the model fails to capture dependencies from the start to the end of the sequence.
What is the most likely reason?

Options:

A) Gradient is not backpropagated through the full sequence.
B) The network is not deep enough.

C) The input embeddings are too small.
D) The model is not trained for enough epochs.

Correct Answer: A

Question: 11

Consider an RNN with inputs x_1, x_2, x_3 and corresponding states s_1, s_2, s_3. The state update is $s_i = \sigma(Ux_i + Ws_{i-1} + b)$.
Which parameters are shared across timesteps and updated during BPTT?

Options:

A) U (input-to-hidden weights)
B) W (hidden-to-hidden recurrent weights)
C) Separate W for each timestep
D) b (biases)

Correct Answer: A, B, D

Question: 12

While backpropagating through time, you notice gradients become very small after many timesteps.
What problem is this a classic example of, and how does it affect training?

Options:

A) Exploding gradients causing unstable updates
B) Vanishing gradients making it difficult to learn long-term dependencies
C) Parameter sharing leading to overfitting
D) Loss summation being incorrect

Correct Answer: B

Question: 13

In BPTT, the derivative $\frac{\partial L_t}{\partial W}$ involves a summation over timesteps $\frac{\partial L_t}{\partial s_t} \times \frac{\partial s_t}{\partial s_k} \times \frac{\partial s_k}{\partial W}$.
What does the term $\frac{\partial s_t}{\partial s_k}$ represent?
Options:

A) Gradient of state at timestep t w.r.t. input at timestep k
B) Impact of state at timestep k on state at timestep t (with $k < t$)
C) Loss at timestep k
D) Output weight gradient

Correct Answer: B

Question: 14

You compute the product of Jacobian matrices corresponding to the partial derivatives of states w.r.t. previous states at each timestep. You notice the product's norm grows exponentially large.
What consequence does this have during training?

Options:

- A) Vanishing gradient problem, no gradient flow to earlier timesteps
- B) Exploding gradient problem, which can cause numerical instability
- C) Perfect gradient flow, leading to efficient training
- D) Parameter sharing across timesteps is broken

Correct Answer: B

Question: 15

While evaluating gradients in vanilla RNN across 50 timesteps, you find the gradients for inputs at timestep 1 are nearly zero.
What practical impact does this have on training?

Options:

- A) Early inputs effectively do not learn influence on outputs
- B) Network updates stay stable
- C) Gradients explode at later timesteps
- D) Training becomes faster

Correct Answer: A

Question: 16

You observe that LSTMs solve the vanishing gradient problem better than vanilla RNNs because:

Options:

- A) Gradients flow freely through the cell state controlled by gates
- B) LSTMs have no gates
- C) Gradients vanish exponentially in LSTMs as well
- D) LSTMs ignore previous states

Correct Answer: A

Question: 17

Despite solving vanishing gradients, LSTMs can still face exploding gradients. Why?

Options:

A) Because gating cannot stop gradient magnitude from growing
B) Because LSTMs lack parameters
C) Because gradients never grow in LSTMs
D) Because of lack of backpropagation

Correct Answer: A

Question: 18

To mitigate exploding gradients in LSTM training, a common practical technique is:

Options:

A) Gradient clipping
B) Increasing learning rate
C) Removing gates
D) Ignoring gradients

Correct Answer: A

Question: 19

A GRU update gate controls:

Options:

A) How much of the previous hidden state to keep
B) How much of the current input is ignored
C) How much to reset the memory
D) How much to forget the output

Correct Answer: A

Question: 20

You are designing a GRU for a language task. It is said to have fewer gates than an LSTM. What is the primary structural difference between a GRU and an LSTM?

Options:

A) GRUs have a separate forget gate while LSTMs do not
B) GRUs combine the forget and input gates into an update gate
C) GRUs do not use gating at all
D) LSTMs use reset and update gates

Correct Answer: B

Question: 21

The GRU uses a reset gate while LSTMs do not. The reset gate in a GRU:

Options:

A) Controls how much past information to forget for the current candidate state
B) Controls the output of the network
C) Is equivalent to the output gate in LSTMs
D) Has no role in state update

Correct Answer: A

Question: 22

In a GRU, the state update equation involves:

Options:

A) A weighted sum of the previous state and new candidate state controlled by the update gate
B) An explicit forget gate separate from the input gate
C) Reset gate modulating the previous state before candidate computation
D) Output gate controlling hidden state visibility

Correct Answers: A, C

Question: 23

The sigmoid activation in gates of LSTMs and GRUs ensures:

Options:

A) Gate values are between 0 and 1 to act as soft selectors
B) Gate outputs are unrestricted real numbers
C) Gates output only binary 0 or 1
D) Activations are always 0

Correct Answer: A

Question: 24

In an LSTM cell, the forget gate outputs values close to 0 for the cell state at time $t-1$. What is the implication?

Options:

A) Cell retains full memory.
B) Cell erases most of the old information.
C) Gradient flow is enhanced.
D) Gradient flow is diminished.

Correct Answers: B, D

Question: 25

A model is predicting POS tags for the sentence:

"Learning deep networks is challenging."

At the word "challenging", the predicted tag is incorrect. The training uses vanilla RNNs.

What might improve performance?

Options:

A) Replacing RNN with LSTM.
B) Increasing learning rate.
C) Increasing hidden state size.
D) Including future context via bidirectional RNNs.

Correct Answers: A, C, D

Question: 26

A sequential model is trained to predict next-day stock movement based on the past 30 days of data. However, due to noisy inputs, the model starts overfitting and performs poorly on unseen sequences.
Which of the following would help regularize the model and improve generalization?

Options:

A) Add dropout between LSTM layers
B) Increase hidden units in LSTM
C) Reduce the number of input time steps
D) Use LSTM with layer normalization

Correct Answers: A, D

Question: 27

You observe that an LSTM is producing almost the same output at every time step despite having different inputs.

Which of the following are likely causes?

Options:

A) Forget gate outputs are close to 1 always
B) Output gate values are near 0
C) Input gate values are near 0
D) Reset gate is not functioning

Correct Answers: B, C

Question: 28

You want to train a model for music generation where the next note depends on a very long history of previous notes. Which model is most appropriate?

Options:

A) Vanilla RNN
B) LSTM
C) GRU
D) Feedforward Neural Network

Correct Answers: B, C

Question: 29

A GRU-based model performs well during early training but eventually diverges, with increasing loss and large weight updates.

What is the most likely cause?

Options:

A) Gradient clipping was not used
B) Reset gate is always zero
C) Learning rate is too small
D) Reset gate is always one

Correct Answer: A

Question: 30

Consider an LSTM with cell state $C_t = 6.0$ at time t, and output gate value $o_t = 0.5$. The hidden state is computed as: $h_t = o_t \, . \, tanh(C_t)$

What is the value of h_t approximately?

Options:

A) 0.5
B) 0.99
C) 1.5
D) 2.5

Correct Answer: A